

Arko Datta

PhD Scholar at SeNSE Lab, IIT Madras

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EDUCATION

Indian Institute of Technology, Madras

Ph.D in Computer Science and Engineering

Sensing and Networked Systems Engineering (SeNSE) Lab, guide: Dr. Ayon Chakraborty

Chennai, Tamil Nadu, India

Jul 2025 (Ongoing)

National Institute of Technology, Durgapur

Masters of Technology in Computer Science and Engineering; GPA: 9.60/10.00

Systems Research Lab (SRL), guide: Dr. Sujoy Saha

Durgapur, West Bengal, India

Aug 2023 – May 2025

Kalinga Institute of Industrial Technology

Bachelor of Technology in Computer Science and Engineering; GPA: 9.31/10.00

Bhubaneswar, Odisha, India

Jul 2018 – Jul 2022

M.P. Birla Foundation Higher Secondary School

ISC (Class XII); Score: 92.6%

Kolkata, West Bengal, India

Apr 2018

AWARDS & ACHIEVEMENTS

Institute Gold Medal: For achieving the highest CGPA in the Department of Computer Science and Engineering, in NIT Durgapur. (Dec 2025)

Institute Best Project: Awarded as the best work among all postgraduate students across departments, in NIT Durgapur. Awarded for I-VayuWise: M.Tech. thesis, which provides an end-to-end intelligent air quality monitoring, mitigation and validation system. (Dec 2025)

Travel Award: Awarded student travel award to present our research work at COMSNETS 2025, held at Chancery Pavilion Hotel, Bengaluru. (Jan 2025)

Second Place, Best Student Idea: Received by our team, "Team PolluSense", in the Student Design Contest, ISSS International Conference on Micro, Nano, and Smart Systems (IC-MNSS 2024). Our team comprised of myself and Pabitra Majhi, hailing from Systems Research Lab, NIT Durgapur and guided by Dr. Sujoy Saha. (Jul 2024)

Travel Award: Awarded student travel award to present our research work at ICDCN 2024, held at IIT Madras. (Jan 2024)

PUBLICATIONS

- **ARGOS: Leveraging Visual Priors for Scalable Wireless Navigation in Dynamic Environments**
Arko Datta, Tharaneeshwaran V U, Aravindh Sriram Kumar A G, and Ayon Chakraborty, PERCOM 2026
- **Adaptive Threshold Learning for Robust First-Path Detection in UWB Ranging**
Shrishti Bekal Mudiakal, Akash Kumar Paul, Arko Datta, Ayon Chakraborty, COMSNETS 2026
- **iAirGuard: A Modular IoT Architecture with Dynamic Sampling and TinyML-Based Fault Detection for Air Quality Monitoring**
Molay Mondal, Arko Datta, Prashanth Durgam, Astom Mondal, Khushi Shyam, Khushi Shyam, Subrata Nandi, Sanghita Bhattacharjee and Sujoy Saha, COMSNETS 2026
- **PolluSenseCheck: Cost Effective TinyML-based Air Quality Monitoring System with In-Built Sensor Fault Detection**
Arko Datta, Joyita Chakraborty, Pabitra Majhi, Astom Mondal, Satyaki Roy, Subrata Nandi and Sujoy Saha, COMSNETS 2025
- **Efficient Air Quality Index Prediction on Resource-Constrained Devices using TinyML: Design, Implementation, and Evaluation**
Arko Datta, Aniruddha Pal, Ranbir Marandi, Nilanjan Chattaraj, Subrata Nandi and Sujoy Saha, ICDCN 2024
- **Real-Time Air Quality Predictions for Smart Cities using TinyML**
Arko Datta, Aniruddha Pal, Ranbir Marandi, Nilanjan Chattaraj, Subrata Nandi and Sujoy Saha, ICDCN 2024

PROJECTS

Generative Signal Maps with Visual Priors | IIT Madras

July 2025 – Present

- A project granted by ANRF ARG to aid scalable wireless sensing deployments in dynamic scenarios, with the aid of visual priors
- **Publications:** PERCOM '26, COMSNETS '26

iPolluSense | NIT Durgapur

March 2024 – May 2025

- A project funded by AI4ICPS IIT Kharagpur, to develop intelligent air and noise quality monitoring device that provides personalised insights for the user's pollution exposure, as well as perform self-diagnosis and correction of sensors.
- **Publications:** COMSNETS '25, COMSNETS '26

IntAirSense: Intelligent Air Pollution Monitoring | NIT Durgapur

September 2023 – March 2024

- A project funded by West Bengal DST, to develop Development of air pollution monitoring systems which use machine learning to predict a more granular AQI.
- **Publications:** ICDCN '24 (2 publications)

EXPERIENCE

Indian Institute of Technology Madras

Chennai, Tamil Nadu, India

Teaching Assistant

January 2026 – Present

- Assisted in delivering lectures for a batch of 50 professional M.Tech. students from notable companies such as Qualcomm, Cert-In and MeITY
- Enhanced learning experience of students via demos and evaluation materials

Indian Institute of Technology Madras

Chennai, Tamil Nadu, India

Web Development Assistant

July 2025 – Present

- Revamped and maintained the website for the Department of Computer Science and Engineering of IIT Madras.

National Institute of Technology, Durgapur

Durgapur, West Bengal, India

Teaching Assistant

August 2023 – May 2025

- Assisted in delivering lectures and perform laboratory activities in the Data Structures, Computer Networks, Compiler Design, and Data Science laboratories.
- Prepared presentations and teaching materials to supplement Data Structures, Data Science, Internet of Things, and Computer Networks lectures.

Indian Institute of Technology, Madras

Chennai, Tamil Nadu, India

Summer Researcher

May 2024 – July 2024

- Was involved in a research on Advanced Capacitive Sensing in the Sensing and Networked Systems Engineering (SENSE) lab at IIT Madras, under Dr. Ayon Chakraborty, department of CSE.

HighRadius Technologies Pvt. Ltd.

Bhubaneswar, Odisha, India

Associate Software Engineer - I

July 2022 – March 2023

- Developed product configuration migration solutions to reduce configuration migration process by 90%.
- Developed product configuration solutions to reduce configuration process by 70%.
- Trained and mentored a batch of hundred interns to develop configuration and migration solutions.

HighRadius Technologies Pvt. Ltd.

Remote

Full-Time Intern

April 2021 – July 2022

- Developed product configuration migration solutions to reduce configuration migration process by 90%.
- Developed product configuration solutions to reduce configuration process by 70%.

HighRadius Technologies Pvt. Ltd.

Remote

Part-Time Intern

January 2021 – March 2021

- Received training in the field of Machine Learning and full stack web development using ReactJS and Java.
- Developed a full stack AI-enabled B2B Order Management System.

SKILLS

Languages: Python, C/C++, Java, JavaScript

Technical Skills: Systems Development, Firmware Development, Computer Vision, Deep Learning

Soft Skills: Leadership, Teamwork, Quick-Learner, Eye for Detail

Hobbies: Piano, Harmonica, Kalimba